# Ganesh Gupta

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## Experience

**Oracle** | *Software Engineer, AI Platform* Jun 2025 – Present

- Led a production **multi-agent system** on **OpenAI Agents SDK**, **LangGraph**, and **MCP** tool use that autonomously processes **financial workflows**, handling **millions of daily transactions** at **99%+ uptime**.
- Architected **GPU/HPC infrastructure** for AI workloads: **CUDA**-based compute clusters with **InfiniBand** interconnects and **Slurm** for **distributed training**.
- Built **evaluation harnesses** measuring model output quality, safety, and **hallucination rates** across versions, gating every release with **regression suites**.
- Shipped **SSE-based streaming** agent delivery with **guardrails** and real-time **observability**, cutting redundant API calls **90%**.

**QureAI** | *AI Engineer* Mar 2026 – Present

- Built **multi-modal computer vision** models on **CNN** and **ResNet** backbones with **UNet segmentation** in **PyTorch** for medical image understanding.
- Trained and fine-tuned vision models on **multi-GPU** clusters, applying **model compression** and **quantization** for efficient production inference.
- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** and **re-ranking**.

**Nomura Research Institute Financial Technologies** | *Software Engineer* Sep 2019 – Aug 2023

- Architected and operated **25+ production Python and Java microservices** for **1M+ users** in **financial services** at **99.9% uptime**.
- Improved vision model accuracy **9.6 points** and cut false negatives **31%** by evaluating **CNN, ResNet, UNet, and ViT** architectures and tuning **augmentation** and thresholds.
- Increased inference throughput **2.4x** and cut **GPU memory** use **27%** by optimizing **batching** and concurrency in **Python** model serving.
- Cut production escalations **60%** in one quarter by standardizing API contracts and failure handling, driving **root cause analysis** and post-incident reviews.

## Projects

**Road Lane Detection (M.S. Thesis, UT Arlington)** `PyTorch`

- Built and benchmarked **lane segmentation** models for road scenes, comparing **UNet**, **CNN**, **Vision Transformer (ViT)**, and **Swin Transformer** architectures in **PyTorch**.

**HoG-ViT** `github.com/ganeshhgupta/HoG-ViT`

- **Vision Transformer** that folds **Histogram of Oriented Gradients (HOG)** features into patch embeddings for **image classification**.

**RootCause AI** `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that routes production alerts to specialized sub-agents via **tool-use** chains and writes structured root-cause reports, cutting **MTTR 65%**.

## Education

**University of Texas at Arlington** | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

**MCKV Institute of Engineering** | *B.Tech Computer Science* Aug 2016 – Jul 2020

## Skills

**Languages:** Python, Java, SQL, TypeScript

**ML and Perception:** PyTorch, Computer Vision, Supervised Learning, Segmentation, CNN, UNet, ViT, Swin

**Training and Deployment:** Multi-GPU Training, CUDA, TensorRT, Quantization, Model Compression

**MLOps and Infra:** Evaluation Pipelines, Model Serving, Slurm, Docker, Kubernetes, AWS, CI/CD

**Certifications:** AWS Certified Developer Associate, GCP Professional MLOps Engineer